

Assignment 2: **Text Classification (LSTM, Pre-trained word embedding)**

For this assignment, please use Sentiment140 dataset (<https://www.kaggle.com/datasets/kazanova/sentiment140>) and perform following tasks,

load the dataset and display number of samples for each class label (0=neg, 1=neutral, 4=pos)

clean it to remove stop-words and punctuation

split dataset in train and test with 30-70 ratio, ensure stratified split

train following models while validating each of these on 20% of the training data (for 1-3, use a fully-connected model with two hidden layers with dim of 32 and 16 respectively)

using TextVectorization with one-gram multi_hot encoding, keeping max feature size = 10000

using TextVectorization with two-gram multi_hot encoding keeping max feature size = 10000

using TextVectorization with two-gram tf_idf encoding keeping max feature size = 10000

use TextVectorization with max length 140 and max token = 10000 and output mode = int. Learn the embedding through embedding layer and use LSTM with output dim=32

repeat 4, this time use GloVe instead of learning word-embedding through embedding layer

repeat 5, use GloVe-twitter instead of GloVe

repeat 5, replace GloVe with FastText

For each of the above 7 experiments, create a table to report accuracy, precision, recall and F1 scores on test dataset having 30% data of Sentiment140. Discuss which model performs best and why?

How to submit?

Download your python script file and upload it on assignment link before the deadline